Moises Mata

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Education

University of Southern California ..... Los Angeles, CA
Ph.D. in Aerospace Engineering, advised by Dr. Keenan Albee ..... Aug 2026 – Present

Columbia University ..... New York, NY
B.A. in Computer Science and Applied Mathematics ..... Sept 2022 – May 2026

Honors & Awards

NSF Graduate Research Fellowship Program, Honorable Mention 2026
Adaptive Planning and Control for Heterogeneous Multi-Agent Planetary Exploration

Publications

Mahajan, Ishaan, Khai Nguyen, Sam Schoedel, Elakhya Nedumaran, **Moises Mata**, Brian Plancher, and Zachary Manchester (2026). "Code Generation and Conic Constraints for Model-Predictive Control on Microcontrollers with Conic-TinyMPC". In: *IEEE International Conference on Robotics and Automation (ICRA)*. arXiv: 2403.18149 [cs.RO]. URL: https://arxiv.org/abs/2403.18149.

Research & Work Experience

NASA Jet Propulsion Laboratory, FSW Architecture & Infrastructure ..... Pasadena, CA
Intern, Flight Software Engineering ..... Jun 2026 – Aug 2026

– Developed long-term "soak-test" infrastructure for the F Prime flight software framework, enabling extended-duration reliability and endurance testing.

Barnard/Dartmouth Accessible and Accelerated Robotics Lab ..... New York, NY
Undergraduate Researcher ..... Jan 2025 – May 2026

– Diagnosed hardware constraints and modernized the **C++** codebase for TinyMPC, an open-source convex Model Predictive Control solver designed for resource-constrained platforms. Advising professor: Brian Plancher.

– Implemented **meta-learning** algorithms to enable online controller optimization and adaptation across systems with different dynamics.

Columbia SNL Lab ..... New York, NY
Undergraduate Researcher ..... Sept 2025 – May 2026

– Developed a high-assurance, real-time runtime for spacecraft flight software using **eBPF**, **F Prime**, and **C/C++**, enabling dynamic multithreaded execution of mission logic with strict safety and millisecond deadline guarantees. Advising Professor: Junfeng Yang. (**Publication In Progress**)

NASA Jet Propulsion Laboratory, Small Scale Flight Software Group ..... Pasadena, CA
Intern, Flight Software Engineering ..... May 2025 – Aug 2025

– Developed core infrastructure for the F Prime flight software framework (v4.0.0 release).

– Architected and implemented reusable software stacks (Communications, Command and Data Handling, Filesystem Support, Data Products), reducing setup time for new missions.

NASA Jet Propulsion Laboratory, Exoplanet Discovery & Science ..... Pasadena, CA
Undergraduate Research Fellow ..... May 2024 – Aug 2024

– Trained binary classification models using **Python**, **scikit-learn**, within **Jupyter Notebook** to predict the presence of habitable planets from simulated observational data.

Leadership Experience

Columbia Space Initiative ..... New York, NY
Executive Board: Co-President (2025–26), Treasurer (2024–25) ..... March 2024 – May 2026

– Led **Columbia's largest engineering club** (**250+ active members**, **$200k annual budget**) as **Co-President**, coordinating strategy across **13 active projects**.

– Organized major events including company visits, faculty talks, and astronaut visits, engaging Columbia and the broader NYC community in aerospace.

– Conducted career development workshops (resume/CV, internships) for Columbia's undergraduate engineering population and led external outreach to **1000+ middle and high school students**, primarily translating engineering concepts into Spanish to inspire underrepresented demographics.

Projects

Flight Software Lead, LIONESS (6U CubeSat) ..... Aug 2024 – May 2026

– Scientific mission to image the circumgalactic medium of nearby galaxies in H-alpha spectrum. Spectrograph instrument developed with Schiminovich Astronomy Instrumentation Lab, Advised by Professor David Schiminovich.

– Architected flight software infrastructure for a science-driven mission. Co-authored NASA Astrophysics Research and Analysis proposal (rated **"Selectable"**); manifested via **NASA CubeSat Launch Initiative**.

Flight Software Lead, PROVES Alcyone (1U CubeSat) ..... Aug 2024 – May 2026

– **Columbia's first satellite**, launched April 2026 to capture Earth imagery. Implemented **F Prime** components for watchdog, power management, and offboard camera control via UART.

NASA SUITS Mission Co-Lead ..... Sept 2023 – Jul 2024

– Co-authored proposal selected as one of **17 national finalists**; developed an AR astronaut-assistance interface in **Unity** (**C#**) for **Microsoft HoloLens 2** and led testing at **NASA Johnson Space Center**.

Skills

Languages	C, C++, C#, Python, MATLAB, Julia
Frameworks & Tools	F Prime, Unity, scikit-learn, CircuitPython, Jupyter
Focus Areas	Robotic Optimization & Control, Embedded Systems, Flight Software
Other	English (native), Spanish (native), French (limited); Amateur Radio (Technician)